

PAPER_1610 — Fe-56 Binding Energy per A = Fe-56 BE/A = $F \cdot K^5 - \beta^4 + 5 \approx 8.792 \text{ MeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Nuclear **Date:** June 18, 2026 **Status:** CLOSED — 0.025% residual

Observation

PAPER_1209II_S663 (Nuclear Unified Proof Set) gives the closed-form expression:

$$\text{Fe-56 BE/A} = F \cdot K^5 - \beta^4 + 5 \approx 8.792 \text{ MeV}$$

Residual to AME2020: **0.025%**.

UQFF Closed Identity

$$\text{Fe-56 BE/A} = F \cdot K^5 - \beta^4 + 5 \approx 8.792 \text{ MeV} \quad 0.025\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S663
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_fe56_be_a_8_7903"})`

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